

An Explicit Evaluation of a Fibonacci Character Sum for Primes with Full Rank of Apparition*

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Abstract

Let $p \geq 7$ be a prime, let χ_p denote the quadratic character modulo p , and let (F_n) be the Fibonacci sequence. We evaluate the character sum

$$S(p) = \sum_{n=1}^{p-1} \chi_p(F_n)$$

under the hypothesis $\alpha(p) = p - 1$, where $\alpha(p)$ denotes the rank of apparition of p in the Fibonacci sequence.

The proof identifies a structural mechanism specific to this full-rank regime. The hypothesis forces $x^2 - x - 1$ to split over $\mathbb{F}_p$, implies $p \equiv 3 \pmod{4}$, and makes the nonresidue root of $x^2 - x - 1$ a primitive root of $\mathbb{F}_p^\times$. This yields a complete multiplicative parametrization of the Fibonacci indices and reduces the original sum to the single character value

$$S(p) = \chi_p(r_- - r_+),$$

where r_- and r_+ are respectively the nonresidue and residue roots of $x^2 - x - 1$. A cyclotomic argument involving fifth roots of unity then gives

$$S(p) = \begin{cases} +1, & p \equiv 11 \pmod{20}, \\ -1, & p \equiv 19 \pmod{20}. \end{cases}$$

The condition $\alpha(p) = p - 1$ may be viewed as a Fibonacci analogue of an Artin-type primitivity condition. Independent computations agree with the formula for all primes $p \leq 2 \times 10^6$ satisfying $\alpha(p) = p - 1$.

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1. Introduction

The arithmetic of the Fibonacci sequence modulo primes is organized around two classical invariants: the *Pisano period* $\pi(p)$ and the *rank of apparition*

$$\alpha(p) = \min\{m \geq 1 : p \mid F_m\}.$$

Beginning with Lucas [1] and continuing through Wall [2], Robinson [3], and Vinson [4], a central theme has been the relationship between these invariants and the multiplicative structure of finite fields. For primes $p \geq 7$, one has

$$\alpha(p) \mid p - \chi_p(5).$$

The extremal value $\alpha(p) = p - 1$ can occur only in the split case $\chi_p(5) = 1$, which is equivalent to $p \equiv \pm 1 \pmod{5}$. This extremal situation will be called the *full-rank regime* throughout the paper. Rather than studying the period or rank themselves, we investigate the quadratic character sum

$$S(p) = \sum_{n=1}^{p-1} \chi_p(F_n)$$

and ask whether the extremal condition $\alpha(p) = p - 1$ imposes a rigid structure on this sum.

At first sight there is little reason to expect such rigidity. Fibonacci congruences, primitive-root questions, and quadratic-residuosity phenomena have been studied in several related settings [5, 6, 7]. The principal observation of the present paper is that the full-rank hypothesis imposes substantially more structure than merely specifying the value of $\alpha(p)$.

Indeed, the hypothesis produces a primitive-root parametrization of the Fibonacci sequence. More precisely, the quadratic nonresidue root of $x^2 - x - 1$ is shown to be a primitive root of $\mathbb{F}_p^\times$. Once this is established, the Fibonacci index becomes a parametrization of the entire multiplicative group $\mathbb{F}_p^\times$, reducing the original character sum to a single quadratic character value.

Thus the proof follows the structural chain

$$\begin{array}{l}
\alpha(p) = p - 1 \\
\implies \text{primitive-root parametrization} \\
\implies \text{structural reduction} \\
\implies S(p) = \chi_p(r_- - r_+) \\
\implies S(p) = \pm 1.
\end{array} \tag{1.1}$$

From this perspective, the explicit value of the character sum is not the starting point of the argument but its final consequence. The central contribution is the structural mechanism that makes the evaluation possible.

Our main theorem is as follows.

Theorem 1.1 (Main Theorem). *Let $p \geq 7$ be a prime satisfying*

$$\alpha(p) = p - 1.$$

Then

$$p \equiv 11 \pmod{20} \quad \text{or} \quad p \equiv 19 \pmod{20},$$

and

$$S(p) = \begin{cases} +1, & p \equiv 11 \pmod{20}, \\ -1, & p \equiv 19 \pmod{20}. \end{cases}$$

The proof combines three ingredients. First, the full-rank condition forces the nonresidue root of the Fibonacci polynomial to be a primitive root modulo p (Lemma 3.2). Second, this primitive-root parametrization converts the Fibonacci sequence into a complete enumeration of the multiplicative group, causing the character sum to reduce to $\chi_p(r_- - r_+)$ (Proposition 5.1). Finally, a cyclotomic discriminant criterion involving fifth roots of unity determines this remaining character explicitly (Lemma 6.3).

This viewpoint places the paper at the intersection of three classical themes: the arithmetic of Fibonacci sequences, primitive-root phenomena of Artin type, and quadratic character sums. A detailed comparison with prior work, including the precise relationship to the classical results of Shanks, De Leon, and Vinson, is given in the following subsection.

1.1. Relation to the literature

The arithmetic of Fibonacci and Lucas sequences modulo primes has been studied from several complementary viewpoints; standard references include [8, 9, 10, 11]. Lucas [1] established basic divisibility properties of Fibonacci numbers modulo primes, while Wall [2] initiated the systematic study of Pisano periods. The relationship between the Pisano period $\pi(p)$ and the rank of apparition $\alpha(p)$ was investigated further by Robinson [3] and Vinson [4].

Vinson [4, Theorem 4(i)] proved that

$$\frac{\pi(p)}{\alpha(p)} = 1$$

for primes $p \equiv 11, 19 \pmod{20}$. Thus, in these residue classes, the period and rank coincide. This result concerns the period–rank quotient and does not evaluate the character sum considered here. The present work concerns the distinct extremal condition $\alpha(p) = p - 1$. Its objective is not to determine the relationship between $\pi(p)$ and $\alpha(p)$, but to evaluate the character sum $S(p)$ under this rank-maximality hypothesis.

The primitive-root viewpoint adopted here is related to the notion of a *Fibonacci primitive root*, introduced by Shanks [12] and developed further by Shanks–Taylor [13] and De Leon [14]. De Leon [14, Theorem 1] proved that, for $p > 5$, a Fibonacci primitive root exists if and only if

$$p \equiv 1, 9 \pmod{10} \quad \text{and} \quad \pi(p) = p - 1.$$

In De Leon’s notation, $A(p)$ denotes the Pisano period. This period-maximality condition is distinct from the rank-maximality condition $\alpha(p) = p - 1$ considered in the present paper. In general, $\pi(p) = p - 1$ and $\alpha(p) = p - 1$ are not equivalent: for example, $p = 41$ has $\pi(41) = 40$ and admits Fibonacci primitive roots modulo 41, whereas $\alpha(41) = 20$. A precise comparison between the full-rank condition $\alpha(p) = p - 1$ and the Fibonacci-primitive-root condition, restricted to the split case, is given in [Proposition 3.4](#) below.

The splitting behaviour of $x^2 - x - 1$ over finite fields and further structural aspects of ranks and periods are considered in, among other works, [15, 16, 17, 18, 19, 20]; the limiting density of Fibonacci residues modulo prime powers has recently been determined in [21]. Related congruence and quadratic-residuosity questions for Fibonacci and Lucas sequences appear in [5, 6, 22, 23].

The principal novelty of the present paper is therefore not the classical theory of Fibonacci primitive roots or period-rank relations. Rather, it is the structural reduction

$$S(p) = \chi_p(r_- - r_+),$$

obtained from the primitive-root parametrization forced by $\alpha(p) = p - 1$, together with the resulting explicit evaluation of this character by a cyclotomic discriminant argument. We are not aware of any previous work establishing either this reduction or the resulting formula for $S(p)$ in the full-rank regime.

The condition $\alpha(p) = p - 1$ also places the problem in an Artin-type setting. For the classical primitive-root problem, see Hooley's conditional density theorem [24] and Moree's survey [25]. The density of primes admitting Fibonacci primitive roots was studied conditionally by Sander [26]. These results concern related, but distinct, primitive-root questions.

1.2. Organisation

Section 2 fixes notation and records elementary facts. Section 3 establishes the primitive-root structure forced by the full-rank hypothesis. Section 4 and Section 5 derive the structural reduction of the character sum. Section 6 determines the remaining sign, and Section 7 proves Theorem 1.1. Computational checks are reported in Section 8, and Section 9 discusses density and generalization questions.

2. Preliminaries

Throughout, p denotes a prime with $p \geq 7$, unless explicitly stated otherwise. We use the standard extension of the quadratic character to all of $\mathbb{F}_p$ by setting $\chi_p(0) = 0$, and recall that

$$\chi_p(-1) = (-1)^{(p-1)/2}.$$

We write

$$\text{QR}_p^\times = \{u \in \mathbb{F}_p^\times : \chi_p(u) = +1\}, \quad \text{NQR}_p^\times = \{u \in \mathbb{F}_p^\times : \chi_p(u) = -1\}.$$

When $p \equiv \pm 1 \pmod{5}$, the polynomial $x^2 - x - 1$ splits over $\mathbb{F}_p$. Fix a square root $s \in \mathbb{F}_p$ of 5, and write its two roots as

$$r_1 = \frac{1+s}{2}, \quad r_2 = \frac{1-s}{2}.$$

Then

$$r_1 + r_2 = 1, \quad r_1 r_2 = -1, \quad r_1 - r_2 = s.$$

The Binet formula holds in $\mathbb{F}_p$:

$$F_m = \frac{r_1^m - r_2^m}{r_1 - r_2}. \quad (2.1)$$

After the quadratic-character analysis in [Lemma 3.2](#), in the full-rank regime we denote the nonresidue root by r_- and the residue root by r_+ :

$$\begin{aligned} \chi_p(r_-) &= -1, & \chi_p(r_+) &= +1, \\ r_- + r_+ &= 1, & r_- r_+ &= -1. \end{aligned}$$

The Binet formula may then be written as

$$F_m = \frac{r_-^m - r_+^m}{r_- - r_+}.$$

In [Section 6](#), the choice of s is additionally made so that $\chi_p(s) = +1$. This choice is independent of the labels $r_{\pm}$, which are determined by quadratic character.

Elementary group-theoretic facts

Lemma 2.1 (Quadratic character in a cyclic group). *Let g be a primitive root of $\mathbb{F}_p^\times$. Then*

$$\chi_p(g^k) = (-1)^k$$

for every integer k .

Proof. The nonzero squares in $\mathbb{F}_p^\times$ are precisely the even powers of g . Since $p - 1$ is even, the parity of k is well defined modulo $p - 1$. ■

Lemma 2.2 (Primitive-root reindexing). *Let g be a primitive root of $\mathbb{F}_p^\times$. Then the map*

$$n \longmapsto g^n$$

from $\{1, \dots, p - 1\}$ to $\mathbb{F}_p^\times$ is a bijection. Consequently, for every function $H : \mathbb{F}_p^\times \rightarrow A$, where A is an abelian group,

$$\sum_{n=1}^{p-1} H(g^n) = \sum_{u \in \mathbb{F}_p^\times} H(u).$$

Moreover,

$$\chi_p(g^n) = (-1)^n \quad (1 \leq n \leq p - 1).$$

Proof. Since g has order $p - 1$, the powers $g, g^2, \dots, g^{p-1}$ are exactly the elements of $\mathbb{F}_p^\times$, each occurring once. The character formula follows from [Lemma 2.1](#). ■

Lemma 2.3 (Fifth roots and the cyclotomic polynomial). *Let K be a field of characteristic different from 5, and let*

$$\Phi_5(X) = X^4 + X^3 + X^2 + X + 1.$$

For $z \in K$, one has $\Phi_5(z) = 0$ if and only if $z \in K^\times$, $z^5 = 1$, and $z \neq 1$. In particular,

$$\Phi_5(z) = 0 \implies \text{ord}(z) = 5.$$

Proof. The identity

$$X^5 - 1 = (X - 1)\Phi_5(X)$$

shows that $\Phi_5(z) = 0$ implies $z^5 = 1$. Moreover, $\Phi_5(1) = 5 \neq 0$, so $z \neq 1$. Conversely, if $z^5 = 1$ and $z \neq 1$, then

$$(z - 1)\Phi_5(z) = 0$$

and hence $\Phi_5(z) = 0$. ■

3. Full-rank structure

Lemma 3.1 (Rank–order identity). *Let r_1, r_2 be the two roots of $x^2 - x - 1$ in $\mathbb{F}_{p^2}$, and put*

$$q = \frac{r_1}{r_2}.$$

Then

$$\alpha(p) = \text{ord}_{\mathbb{F}_{p^2}^\times}(q).$$

In particular, if $x^2 - x - 1$ splits over $\mathbb{F}_p$, then for each root $r \in \mathbb{F}_p$,

$$\alpha(p) = \text{ord}_{\mathbb{F}_p^\times}(-r^2).$$

Proof. Since $p \neq 5$, the roots are distinct. Binet’s formula in $\mathbb{F}_{p^2}$ gives

$$F_m \equiv 0 \pmod{p} \iff r_1^m = r_2^m \iff q^m = 1.$$

Thus

$$\alpha(p) = \text{ord}_{\mathbb{F}_{p^2}^\times}(q).$$

Now suppose that $x^2 - x - 1$ splits over $\mathbb{F}_p$, and let r be either root. The other root is $-r^{-1}$, since the product of the roots is -1 . Thus the ratio r_1/r_2 is either $-r^2$ or its inverse, depending on the ordering of the roots. These elements have the same multiplicative order, so

$$\alpha(p) = \text{ord}_{\mathbb{F}_p^\times}(-r^2).$$

■

Lemma 3.2 (Primitive-root mechanism in the full-rank regime). *Let $p \geq 7$ be a prime such that*

$$\alpha(p) = p - 1.$$

Then $x^2 - x - 1$ splits over $\mathbb{F}_p$, and

$$p \equiv 3 \pmod{4}.$$

Label its two roots $r_+, r_- \in \mathbb{F}_p^\times$ by their quadratic characters:

$$\chi_p(r_+) = +1, \quad \chi_p(r_-) = -1.$$

Then

$$\text{ord}_{\mathbb{F}_p^\times}(-r_-^2) = p - 1$$

and

$$\text{ord}_{\mathbb{F}_p^\times}(r_-) = p - 1.$$

In particular, r_- is a primitive root of $\mathbb{F}_p^\times$.

Proof. Let r_1, r_2 be the two roots of $x^2 - x - 1$ in $\mathbb{F}_{p^2}$, and put

$$q = \frac{r_1}{r_2}.$$

Suppose first that $x^2 - x - 1$ does not split over $\mathbb{F}_p$. The Frobenius automorphism of $\mathbb{F}_{p^2}$ interchanges its roots, so

$$r_1^p = r_2, \quad r_2^p = r_1.$$

Consequently,

$$q^p = \frac{r_1^p}{r_2^p} = \frac{r_2}{r_1} = q^{-1},$$

and hence

$$q^{p+1} = 1.$$

Thus $\text{ord}(q) \mid p+1$. By [Lemma 3.1](#),

$$\alpha(p) \mid p+1.$$

Since $\alpha(p) = p-1$, this gives $p-1 \mid p+1$, hence $p-1 \mid 2$, contradicting $p \geq 7$. Therefore $x^2 - x - 1$ splits over $\mathbb{F}_p$.

Let r be either root in $\mathbb{F}_p$. By the split-case assertion of [Lemma 3.1](#),

$$\text{ord}(-r^2) = p-1.$$

If $\chi_p(-1) = +1$, then $-r^2$ is a nonzero square and lies in the subgroup of squares of order $(p-1)/2$, contradicting $\text{ord}(-r^2) = p-1$. Hence

$$\chi_p(-1) = -1,$$

or equivalently,

$$p \equiv 3 \pmod{4}.$$

Since $r_1 r_2 = -1$ and $\chi_p(-1) = -1$, the roots have opposite quadratic characters. Hence precisely one root is a residue and the other is a nonresidue; denote them by r_+ and r_- , respectively.

Applying [Lemma 3.1](#) to r_- gives

$$\text{ord}(-r_-^2) = \alpha(p) = p-1.$$

Since $\chi_p(r_-) = -1$, Euler's criterion yields

$$r_-^{(p-1)/2} = -1,$$

so

$$-1 \in \langle r_- \rangle.$$

Because $r_-^2 \in \langle r_- \rangle$, it follows that

$$-r_-^2 \in \langle r_- \rangle.$$

This element has order $p-1$, so $\langle r_- \rangle$ contains an element of order $|\mathbb{F}_p^\times| = p-1$. Therefore

$$\langle r_- \rangle = \mathbb{F}_p^\times,$$

and hence

$$\text{ord}(r_-) = p-1.$$

■

Remark 3.3 (Role of the full-rank hypothesis). The essential consequence of [Lemma 3.2](#) is that the substitution

$$u = r_-^n$$

is a bijection from $\{1, \dots, p-1\}$ onto $\mathbb{F}_p^\times$. This allows the Fibonacci index to be replaced by a multiplicative parameter in the proof of [Proposition 5.1](#).

Proposition 3.4 (Full rank and Fibonacci primitive roots). *Let $p > 5$ be a prime. Then, in the split case $\chi_p(5) = 1$, the following implications hold:*

$$\alpha(p) = p-1 \implies p \equiv 3 \pmod{4}$$

and $x^2 - x - 1$ has a Fibonacci primitive root modulo p . Conversely, if $p \equiv 3 \pmod{4}$ and $x^2 - x - 1$ has a Fibonacci primitive root modulo p , then

$$\alpha(p) = p-1.$$

Proof. The forward implication is the content of [Lemma 3.2](#): in the full-rank regime the nonresidue root r_- of $x^2 - x - 1$ satisfies

$$\text{ord}_{\mathbb{F}_p^\times}(r_-) = p-1,$$

so r_- is a Fibonacci primitive root.

Conversely, let g be a Fibonacci primitive root. The other root of $x^2 - x - 1$ is $-g^{-1}$, and the split-case identity of [Lemma 3.1](#) therefore gives

$$\alpha(p) = \text{ord}_{\mathbb{F}_p^\times}(-g^2).$$

Since $p \equiv 3 \pmod{4}$ and g is primitive,

$$-1 = g^{(p-1)/2},$$

whence

$$-g^2 = g^{(p+3)/2}.$$

Writing $p = 4m + 3$, we have $p-1 = 4m+2$ and $(p+3)/2 = 2m+3$. Hence

$$\gcd\left(p-1, \frac{p+3}{2}\right) = \gcd(4m+2, 2m+3).$$

Any common divisor divides

$$(4m+2) - 2(2m+3) = -4.$$

Since $2m + 3$ is odd, the common divisor is odd and divides 4; therefore it is equal to 1. It follows that

$$\text{ord}_{\mathbb{F}_p^\times}(-g^2) = p - 1,$$

and consequently $\alpha(p) = p - 1$. ■

Remark 3.5 (Relation to Fibonacci primitive roots). [Proposition 3.4](#) makes the comparison with the classical theory of Shanks [12] and De Leon [14] completely explicit: in the split case, full rank is equivalent to the existence of a Fibonacci primitive root together with the congruence $p \equiv 3 \pmod{4}$. The direction used in the body of the paper is the forward implication, which supplies the primitive-root parametrization needed for the character-sum evaluation.

Remark 3.6 (Order of the residue root). In the full-rank regime, put

$$n = \frac{p - 1}{2}.$$

Since $p \equiv 3 \pmod{4}$, the integer n is odd. Since r_- is primitive,

$$r_-^n = -1.$$

Using $r_- r_+ = -1$, we obtain

$$r_+ = -r_-^{-1} = r_-^{n-1}.$$

Writing $p = 4m + 3$, so that $n = 2m + 1$, gives

$$\text{ord}(r_+) = \frac{p - 1}{\gcd(p - 1, n - 1)} = \frac{4m + 2}{\gcd(4m + 2, 2m)} = \frac{4m + 2}{2} = \frac{p - 1}{2}.$$

4. Auxiliary character-sum identities

The following elementary identities isolate the only character sums required in the structural reduction.

Lemma 4.1 (A quadratic character-sum identity). *For every $a \in \mathbb{F}_p^\times$,*

$$\sum_{x \in \mathbb{F}_p} \chi_p(x^2 - a) = -1.$$

Proof. This is a standard quadratic-character-sum identity; see, for example, [11]. We include a proof for completeness.

Put

$$T(a) := \sum_{x \in \mathbb{F}_p} \chi_p(x^2 - a).$$

Since

$$\#\{x \in \mathbb{F}_p : x^2 = t\} = 1 + \chi_p(t),$$

we have

$$T(a) = \sum_{t \in \mathbb{F}_p} (1 + \chi_p(t)) \chi_p(t - a) = \sum_{t \in \mathbb{F}_p} \chi_p(t(t - a)),$$

because

$$\sum_{t \in \mathbb{F}_p} \chi_p(t - a) = 0.$$

The substitution $t = aw$ shows that $T(a)$ is independent of $a \neq 0$. Since

$$T(0) = \sum_{x \in \mathbb{F}_p} \chi_p(x^2) = p - 1$$

and

$$\sum_{a \in \mathbb{F}_p} T(a) = 0,$$

the common value of $T(a)$ for $a \neq 0$ is -1 . ■

Lemma 4.2 (Negation interchanges residues and nonresidues). *If $p \equiv 3 \pmod{4}$, then the map*

$$u \longmapsto -u$$

is a bijection

$$\text{QR}_p^\times \longrightarrow \text{NQR}_p^\times.$$

Proof. For $u \in \mathbb{F}_p^\times$,

$$\chi_p(-u) = \chi_p(-1)\chi_p(u) = -\chi_p(u).$$

The two sets have the same cardinality $(p - 1)/2$. ■

Lemma 4.3 (Restricted quadratic character sums). *If $p \equiv 3 \pmod{4}$, then*

$$\sum_{u \in \text{QR}_p^\times} \chi_p(u^2 - 1) = 0$$

and

$$\sum_{u \in \text{NQR}_p^\times} \chi_p(u^2 + 1) = -1.$$

Proof. By Lemma 4.1 with $a = 1$, and isolating the term $x = 0$, we obtain

$$\sum_{x \in \mathbb{F}_p^\times} \chi_p(x^2 - 1) = -1 - \chi_p(-1) = 0.$$

Since $(-u)^2 - 1 = u^2 - 1$ and negation maps $\text{QR}_p^\times$ bijectively onto $\text{NQR}_p^\times$ by Lemma 4.2,

$$\sum_{u \in \text{QR}_p^\times} \chi_p(u^2 - 1) = \sum_{v \in \text{NQR}_p^\times} \chi_p(v^2 - 1).$$

Thus each of these two sums is zero.

Similarly, Lemma 4.1 with $a = -1$ gives

$$\sum_{x \in \mathbb{F}_p^\times} \chi_p(x^2 + 1) = -1 - \chi_p(1) = -2.$$

Again, because $(-u)^2 + 1 = u^2 + 1$ and negation interchanges the two character classes,

$$\sum_{u \in \text{NQR}_p^\times} \chi_p(u^2 + 1) = \sum_{v \in \text{QR}_p^\times} \chi_p(v^2 + 1).$$

Therefore each of these two sums equals -1 . ■

5. The structural reduction

Proposition 5.1 (Structural reduction of the character sum). *Let $p \geq 7$ satisfy*

$$\alpha(p) = p - 1.$$

Then

$$S(p) = \chi_p(r_- - r_+),$$

where r_- and r_+ are respectively the nonresidue and residue roots of $x^2 - x - 1$ in $\mathbb{F}_p$.

Proof. Put

$$D = r_- - r_+.$$

Since the discriminant of $x^2 - x - 1$ is $5 \neq 0$ in $\mathbb{F}_p$, we have $D \neq 0$. Since

$$r_+ = -r_-^{-1},$$

the Binet formula gives

$$F_n = \frac{r_-^n - (-1)^n r_-^{-n}}{D}.$$

By [Lemma 3.2](#), r_- is a primitive root of $\mathbb{F}_p^\times$. Hence, by [Lemma 2.2](#), the map

$$n \mapsto u = r_-^n$$

is a bijection from $\{1, \dots, p-1\}$ onto $\mathbb{F}_p^\times$, and

$$\chi_p(u) = (-1)^n.$$

Set

$$\varepsilon = \chi_p(D).$$

Then

$$\chi_p(F_n) = \varepsilon \chi_p(u - \chi_p(u)u^{-1}).$$

If $u \in \text{QR}_p^\times$, then $\chi_p(u) = +1$, and the contribution is

$$\varepsilon \chi_p(u^2 - 1).$$

If $u \in \text{NQR}_p^\times$, then $\chi_p(u) = -1$, and the contribution is

$$-\varepsilon \chi_p(u^2 + 1).$$

Therefore, by [Lemma 4.3](#),

$$S(p) = \varepsilon \left(\sum_{u \in \text{QR}_p^\times} \chi_p(u^2 - 1) - \sum_{u \in \text{NQR}_p^\times} \chi_p(u^2 + 1) \right) = \varepsilon(0 - (-1)).$$

Hence

$$S(p) = \varepsilon = \chi_p(r_- - r_+). \quad \blacksquare$$

Remark 5.2. The term $n = p-1$, corresponding to $u = 1$, contributes zero to the sum through the factor $\chi_p(u^2 - 1)$ and therefore requires no separate correction.

6. Evaluation of the remaining sign

Assume throughout this section that

$$p \equiv 3 \pmod{4} \quad \text{and} \quad p \equiv \pm 1 \pmod{5}.$$

Since $\chi_p(5) = 1$, choose $s \in \mathbb{F}_p$ such that

$$s^2 = 5 \quad \text{and} \quad \chi_p(s) = +1.$$

Such a choice is possible because the two square roots of 5 have opposite quadratic characters when $p \equiv 3 \pmod{4}$. Put

$$\phi = \frac{1+s}{2}, \quad \psi = \frac{1-s}{2}.$$

These are the two roots of $x^2 - x - 1$.

Lemma 6.1 (Existence of fifth roots of unity). *There exists $z \in \mathbb{F}_p^\times$ with*

$$z^5 = 1 \quad \text{and} \quad z \neq 1$$

if and only if

$$p \equiv 1 \pmod{5}.$$

Proof. The group $\mathbb{F}_p^\times$ is cyclic of order $p-1$. It contains an element of order 5 if and only if $5 \mid p-1$. ■

Lemma 6.2 (Cyclotomic discriminant criterion). *Let*

$$t = \frac{s-1}{2} \quad \text{and} \quad \Delta = t^2 - 4.$$

Then $\Delta \neq 0$ and

$$\chi_p(\Delta) = \begin{cases} +1, & p \equiv 1 \pmod{5}, \\ -1, & p \equiv -1 \pmod{5}. \end{cases}$$

Proof. Since $s^2 = 5$, one computes

$$t^2 + t - 1 = 0$$

and

$$\Delta = -t - 3.$$

If $\Delta = 0$, then $t = -3$, which together with $t^2 + t - 1 = 0$ gives $5 = 0$ in $\mathbb{F}_p$, contradicting $p \neq 5$. Thus $\Delta \neq 0$.

Suppose first that $\chi_p(\Delta) = +1$. Then the polynomial

$$Z^2 - tZ + 1$$

has a root $z \in \mathbb{F}_p$. Thus

$$z + z^{-1} = t,$$

and hence

$$z^2 + z^{-2} = t^2 - 2.$$

Using $t^2 + t - 1 = 0$, we obtain

$$(z^2 + z^{-2}) + (z + z^{-1}) + 1 = (t^2 - 2) + t + 1 = 0.$$

Multiplying by z^2 yields

$$\Phi_5(z) = 0.$$

By [Lemma 2.3](#),

$$\text{ord}(z) = 5.$$

Therefore $p \equiv 1 \pmod{5}$ by [Lemma 6.1](#).

Conversely, suppose that $p \equiv 1 \pmod{5}$. Choose $z_0 \in \mathbb{F}_p^\times$ of order 5, and put

$$\tau = z_0 + z_0^{-1}.$$

From $\Phi_5(z_0) = 0$, one obtains

$$\tau^2 + \tau - 1 = 0.$$

The two roots of $X^2 + X - 1$ are t and $-1 - t$.

If $\tau = t$, then z_0 satisfies

$$z_0^2 - tz_0 + 1 = 0,$$

so $\chi_p(\Delta) = +1$.

If $\tau = -1 - t$, set $z_1 = z_0^2$. Then z_1 also has order 5, and

$$z_1 + z_1^{-1} = \tau^2 - 2 = 1 - \tau - 2 = -1 - \tau = t.$$

Thus z_1 satisfies

$$z_1^2 - tz_1 + 1 = 0,$$

and again $\chi_p(\Delta) = +1$.

We have proved that

$$\chi_p(\Delta) = +1 \iff p \equiv 1 \pmod{5}.$$

Since $\Delta \neq 0$ and $p \equiv \pm 1 \pmod{5}$, the remaining case gives

$$\chi_p(\Delta) = -1$$

when $p \equiv -1 \pmod{5}$. ■

Lemma 6.3 (Sign determination). *Let r_- and r_+ denote the nonresidue and residue roots of $x^2 - x - 1$, respectively. Then*

$$\chi_p(r_- - r_+) = \begin{cases} +1, & p \equiv 1 \pmod{5}, \\ -1, & p \equiv -1 \pmod{5}. \end{cases}$$

Proof. We have

$$s\phi = s \frac{1+s}{2} = \frac{s+5}{2}.$$

Since

$$\Delta = -t - 3 \quad \text{and} \quad t = \frac{s-1}{2},$$

it follows that

$$\Delta = -\frac{s-1}{2} - 3 = -\frac{s+5}{2} = -s\phi.$$

Taking quadratic characters and using

$$\chi_p(s) = +1, \quad \chi_p(-1) = -1,$$

we obtain

$$\chi_p(\Delta) = -\chi_p(\phi).$$

Suppose that $p \equiv 1 \pmod{5}$. By [Lemma 6.2](#),

$$\chi_p(\Delta) = +1,$$

so

$$\chi_p(\phi) = -1.$$

Thus

$$r_- = \phi, \quad r_+ = \psi,$$

and

$$r_- - r_+ = \phi - \psi = s.$$

Therefore

$$\chi_p(r_- - r_+) = +1.$$

Now suppose that $p \equiv -1 \pmod{5}$. Then

$$\chi_p(\Delta) = -1,$$

so

$$\chi_p(\phi) = +1.$$

Thus

$$r_+ = \phi, \quad r_- = \psi,$$

and

$$r_- - r_+ = \psi - \phi = -s.$$

Therefore

$$\chi_p(r_- - r_+) = \chi_p(-1)\chi_p(s) = -1.$$

■

7. Proof of the main theorem

Proof of Theorem 1.1. By Lemma 3.2, the polynomial $x^2 - x - 1$ splits over $\mathbb{F}_p$ and

$$p \equiv 3 \pmod{4}.$$

Splitting is equivalent to $\chi_p(5) = +1$, hence to

$$p \equiv \pm 1 \pmod{5}.$$

Combining these congruences gives

$$p \equiv 11 \pmod{20} \quad \text{or} \quad p \equiv 19 \pmod{20}.$$

By Proposition 5.1,

$$S(p) = \chi_p(r_- - r_+).$$

By Lemma 6.3, this is $+1$ when $p \equiv 1 \pmod{5}$, equivalently when $p \equiv 11 \pmod{20}$, and it is -1 when $p \equiv -1 \pmod{5}$, equivalently when $p \equiv 19 \pmod{20}$. ■

8. Computational checks

The conclusion of [Theorem 1.1](#) was checked for every prime

$$p \leq 2 \times 10^6$$

satisfying

$$\alpha(p) = p - 1.$$

Table 1: Computational check of [Theorem 1.1](#) for all primes $p \leq 2 \times 10^6$ satisfying $\alpha(p) = p - 1$.

$p \bmod 20$	predicted $S(p)$	count	matches	status
11	+1	11,755	11,755	PASS
19	-1	14,652	14,652	PASS
total		26,407	26,407	PASS

The verification is non-circular: both $\alpha(p)$ and $S(p)$ were computed directly from their definitions, and neither calculation uses the closed formula proved in [Theorem 1.1](#). In particular, $\alpha(p)$ was obtained by iterating the Fibonacci recurrence modulo p and recording the least index $n \geq 1$ for which $F_n \equiv 0 \pmod{p}$, while $S(p)$ was obtained by summing $\chi_p(F_n)$ for $1 \leq n \leq p - 1$. The subsequent consistency checks compare these direct computations with the structural predictions: (i) the quadratic characters of both roots of $x^2 - x - 1$ in $\mathbb{F}_p$ against [Lemma 3.2](#); (ii) the multiplicative orders of $-r_-^2$ and r_- against [Lemmas 3.1](#) and [3.2](#); and (iii) the sign of $\chi_p(r_- - r_+)$ against the discriminant criterion of [Lemma 6.3](#). These checks provide independent computational support for both the predicted value of $S(p)$ and the intermediate identities used in the proof.

Among the 148,933 primes $p \leq 2 \times 10^6$, exactly 26,407 satisfy $\alpha(p) = p - 1$. Among the 74,461 split primes

$$p \equiv \pm 1 \pmod{5},$$

the observed full-rank proportion is

$$\frac{26,407}{74,461} \approx 0.35459 \quad (35.46\%).$$

For comparison, the first-order Artin-type value suggested by the classical classwise Fibonacci-primitive-root densities ([Remark 9.2](#)) is

$$C_F^{\text{heur}} = \frac{18A}{19} \approx 0.35427.$$

The observed ratio differs from this heuristic value by about 3.1×10^{-4} in absolute terms (relative difference below 0.1%). This comparison is empirical only and is not used in the proof of [Theorem 1.1](#). The computation was performed in Python. Source code, datasets, and verification logs are available in the reproducibility archive described in [Section Appendix B](#).

Remark 8.1 (Independent computational cross-check). A separate Python implementation, independent of the main codebase, recomputed $\alpha(p)$ and $S(p)$ for every prime $p \leq 20,000$ and for a random sample of 150 primes satisfying

$$20,000 < p \leq 5 \times 10^5.$$

The independent recomputation agreed with the main computation in every case. Moreover, the proportion

$$\frac{7,379}{20,751} \approx 0.3556$$

among split primes at the bound 5×10^5 , as reported by the main codebase, is consistent with the value 0.35459 obtained at the bound 2×10^6 and with the first-order heuristic $C_F^{\text{heur}} = 18A/19 \approx 0.35427$ of [Remark 9.2](#).

9. Concluding remarks

We have obtained an explicit evaluation of the Fibonacci character sum

$$S(p) = \sum_{n=1}^{p-1} \chi_p(F_n)$$

in the full-rank regime

$$\alpha(p) = p - 1.$$

The proof shows that the sum is governed by the structural reduction

$$S(p) = \chi_p(r_- - r_+),$$

which in turn arises from the primitive-root parametrization forced by the full-rank condition. The remaining sign is determined by $p \bmod 20$.

The split-case identity in [Lemma 3.1](#) identifies $\alpha(p) = p - 1$ with the primitivity of

$$-r^2 \in \mathbb{F}_p^\times$$

for either root r of $x^2 - x - 1$. Thus the full-rank condition belongs naturally to an Artin-type family of primitive-root problems. In the split setting, [Proposition 3.4](#) further identifies the full-rank condition with the Fibonacci-primitive-root condition restricted to the two residue classes $p \equiv 11, 19 \pmod{20}$. This gives the conjectural density constant

$$C_F^{\text{heur}} = \frac{18A}{19}$$

a classical Artin-type interpretation through the classwise Fibonacci-primitive-root densities of Shanks [\[12\]](#).

A density question

Among primes $p \leq 2 \times 10^6$ in the split classes

$$p \equiv \pm 1 \pmod{5},$$

the observed full-rank proportion is

$$\frac{26,407}{74,461} \approx 0.35459$$

(see the numerical verification above). By [Theorem 1.1](#), such primes can occur only in the two classes

$$p \equiv 11, 19 \pmod{20},$$

among the four split residue classes

$$p \equiv 1, 9, 11, 19 \pmod{20}.$$

This motivates the following conjecture.

Conjecture 9.1. *Let*

$$N_F(x) = \#\{p \leq x : \alpha(p) = p - 1\}$$

and

$$N_{\text{split}}(x) = \#\{p \leq x : p \equiv \pm 1 \pmod{5}\}.$$

There exists a constant C_F with

$$0 < C_F \leq \frac{1}{2}$$

such that

$$N_F(x) \sim C_F N_{\text{split}}(x) \quad (x \rightarrow \infty).$$

Equivalently, the natural density of full-rank primes among all primes equals $C_F/2$.

The upper bound

$$C_F \leq \frac{1}{2}$$

is forced by [Theorem 1.1](#). By the split-case identity in [Lemma 3.1](#), [Conjecture 9.1](#) is a primitive-root statement for the reduction of the algebraic unit

$$-\phi^2, \quad \phi = \frac{1 + \sqrt{5}}{2},$$

at primes split in $\mathbb{Q}(\sqrt{5})$. The primitive-root interpretation of [Proposition 3.4](#) gives a classical route to a first-order heuristic for the constant C_F . In the four split classes

$$p \equiv 1, 9, 11, 19 \pmod{20},$$

the full-rank condition can occur only in the two classes

$$p \equiv 11, 19 \pmod{20}$$

by [Theorem 1.1](#). The relevant conditional densities for Fibonacci primitive roots in these two classes were computed class-by-class by Shanks [\[12\]](#); see also the later conditional theorem of Sander [\[26\]](#). Thus the heuristic value of C_F can be obtained directly from the classical Fibonacci-primitive-root densities, after restricting to the two classes allowed by the full-rank theorem.

Remark 9.2 (Heuristic value of C_F). The constant C_F is naturally related to the classical Fibonacci-primitive-root densities. Shanks [\[12\]](#) gives the following first-order densities, conditional on the indicated residue class modulo 20:

$p \bmod 20$	1	9	11	19
$\delta_{\text{FPR}}(p \bmod 20)$	$\frac{16A}{19}$	$\frac{20A}{19}$	$\frac{32A}{19}$	$\frac{40A}{19}$

where

$$A = \prod_q \left(1 - \frac{1}{q(q-1)}\right) \approx 0.3739558136$$

is Artin's constant. The remaining four residue classes modulo 20 have zero density for Fibonacci primitive roots. Shanks's averaging over all eight residue classes yields the classical constant

$$\frac{27A}{38}$$

for the density of primes admitting a Fibonacci primitive root.

For the present problem, however, only the classes $p \equiv 11, 19 \pmod{20}$ can contribute, by [Theorem 1.1](#). Among the four split classes

$$p \equiv 1, 9, 11, 19 \pmod{20},$$

these two classes have asymptotic relative weight $1/2$. Conditional on either of these two classes, the corresponding first-order Fibonacci-primitive-root densities are

$$\frac{32A}{19} \quad \text{and} \quad \frac{40A}{19}.$$

Consequently, the resulting first-order conditional value in [Conjecture 9.1](#) is

$$C_F^{\text{heur}} = \frac{1}{2} \left(\frac{1}{2} \left(\frac{32A}{19} + \frac{40A}{19} \right) \right) = \frac{18A}{19} \approx 0.354274.$$

Thus

$$\boxed{C_F^{\text{heur}} = \frac{18A}{19}}.$$

This value is recorded here as a conjectural first-order constant, not as a proved identity. The relevant analytic problem is an Artin-type primitive-root problem for the algebraic setting arising from the split Fibonacci polynomial. Sander [\[26\]](#) established the classical Fibonacci-primitive-root density under appropriate generalized Riemann hypotheses; a corresponding derivation for the present conditional formulation is not developed here.

It would be natural to ask whether a Hooley-type analysis, under suitable GRH assumptions, yields a rigorous justification of the value $18A/19$. This question lies beyond the scope of the present paper.

Further directions

The same primitive-root parametrization remains available for Lucas-type sums

$$L_n = r_-^n + r_+^n.$$

It is therefore natural to investigate whether an analogue of the reindexing argument in [Proposition 5.1](#) yields explicit evaluations for character sums attached to Lucas sequences; for recent work on the arithmetic of Lucas sequences and the integers they represent, see [\[27\]](#). This extension is not pursued here.

More generally, the present result suggests that extremal rank conditions may provide a mechanism for evaluating character sums attached to linear recurrence sequences. Extending this approach to higher-order characters and to more general Lucas sequences is a natural direction for future work.

Appendix A. Illustrative examples

Table A.2: Illustrative examples of [Theorem 1.1](#) for selected primes p satisfying $\alpha(p) = p - 1$.

p	$\alpha(p)$	$p \bmod 20$	r_-	r_+	$r_- - r_+$
11	10	11	8	4	$4 = 2^2$ (QR)
19	18	19	15	5	10 (NQR)
31	30	11	13	19	$25 = 5^2 \pmod{31}$ (QR)
59	58	19	34	26	8 (NQR)
79	78	19	30	50	59 (NQR)

In each case, the quadratic character of $r_- - r_+$ agrees with [Theorem 1.1](#): the value is $+1$ in the class $11 \pmod{20}$ and -1 in the class $19 \pmod{20}$. Accordingly,

$$\begin{aligned} S(11) &= +1, & S(19) &= -1, & S(31) &= +1, \\ S(59) &= -1, & S(79) &= -1. \end{aligned}$$

Appendix B. Reproducibility archive

The source code, datasets, and verification logs used for the computations are available in the accompanying GitHub repository and archived at [\[28\]](#). The archive includes the principal implementation, `fibchar_v1_0_0.py`, together

with the data files required to reproduce the computational summary in [Table 1](#).

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